COMP 2087: Programming for Beginners II
FALL 2024 LAB #3

Code for both PART 1 and PART 2 should be submitted together in a file *lab3sol.py*

PART 1: Reading from a text file

Upload file *lab3in.txt* into your project directory

Write code to iterate over each line in the file and do the following:

1a. Print the line (without the last newline) and the number of times the character 'e' occurs in that line.

1b. After all the lines have been processed, print the average number of times the character 'e' occurs per line, rounded to 2 decimal places.

Sample output:

```
"C:\Program Files (x86)\python.exe" "C:\Users\ajaek\OneDrive
******* OUTPUT FOR PART 1 ********
line = Whose woods these are I think I know   # of e's = 4
line = His house is in the village though   # of e's = 3
line = He will not see me stopping here   # of e's = 6
line = To watch his woods fill up with snow   # of e's = 0
line = My little horse must think it queer   # of e's = 4
line = To stop without a farmhouse near   # of e's = 2
line = Between the woods and frozen lake   # of e's = 6
line = The darkest evening of the year   # of e's = 6
line = He gives his harness bells a shake   # of e's = 5
line = To ask if there is some mistake   # of e's = 4
line = The only other sound's the sweep   # of e's = 5
line = Of easy wind and downy flake   # of e's = 2
line = The woods are lovely, dark and deep   # of e's = 5
line = But I have promises to keep   # of e's = 4
line = And miles to go before I sleep   # of e's = 5
line = And miles to go before I sleep   # of e's = 5
average # of occurrences of "e" per line = 4.12
```

PART 2: Write additional code

Update the above code to add the following functionality:

2a. For each line in *lab3in.txt* create *list* of words and print the list. This list may contain duplicates. For this program, words are groups of characters separated by one or more spaces.

2b. Add the words in the line to a *set* called ***words***. NOTE: A set is an unordered collection with no duplicate elements. The words are case-sensitive, so for example, 'The' and 'the' will be considered two separate words.

2c. For each line you should also print the contents of the line in *uppercase* in an output file called *lab3out.txt*

2d. After all the lines have been processed, write the final value of ***words*** in the output file.

Sample output (output file): NOTE: screenshot does not show full set of ***words***

```
WHOSE WOODS THESE ARE I THINK I KNOW
HIS HOUSE IS IN THE VILLAGE THOUGH
HE WILL NOT SEE ME STOPPING HERE
TO WATCH HIS WOODS FILL UP WITH SNOW
MY LITTLE HORSE MUST THINK IT QUEER
TO STOP WITHOUT A FARMHOUSE NEAR
BETWEEN THE WOODS AND FROZEN LAKE
THE DARKEST EVENING OF THE YEAR
HE GIVES HIS HARNESS BELLS A SHAKE
TO ASK IF THERE IS SOME MISTAKE
THE ONLY OTHER SOUND'S THE SWEEP
OF EASY WIND AND DOWNY FLAKE
THE WOODS ARE LOVELY, DARK AND DEEP
BUT I HAVE PROMISES TO KEEP
AND MILES TO GO BEFORE I SLEEP
AND MILES TO GO BEFORE I SLEEP
{'And', 'these', 'and', 'sleep', 'snow', 'de
```

Sample output (on screen portion only):

```
******* OUTPUT FOR PART 2 ********
The words in this line: ['Whose', 'woods', 'these', 'are', 'I', 'think', 'I', 'know']
The words in this line: ['His', 'house', 'is', 'in', 'the', 'village', 'though']
The words in this line: ['He', 'will', 'not', 'see', 'me', 'stopping', 'here']
The words in this line: ['To', 'watch', 'his', 'woods', 'fill', 'up', 'with', 'snow']
The words in this line: ['My', 'little', 'horse', 'must', 'think', 'it', 'queer']
The words in this line: ['To', 'stop', 'without', 'a', 'farmhouse', 'near']
The words in this line: ['Between', 'the', 'woods', 'and', 'frozen', 'lake']
The words in this line: ['The', 'darkest', 'evening', 'of', 'the', 'year']
The words in this line: ['He', 'gives', 'his', 'harness', 'bells', 'a', 'shake']
The words in this line: ['To', 'ask', 'if', 'there', 'is', 'some', 'mistake']
The words in this line: ['The', 'only', 'other', "sound's", 'the', 'sweep']
The words in this line: ['Of', 'easy', 'wind', 'and', 'downy', 'flake']
The words in this line: ['The', 'woods', 'are', 'lovely,', 'dark', 'and', 'deep']
The words in this line: ['But', 'I', 'have', 'promises', 'to', 'keep']
The words in this line: ['And', 'miles', 'to', 'go', 'before', 'I', 'sleep']
The words in this line: ['And', 'miles', 'to', 'go', 'before', 'I', 'sleep']
```

Submit your Python code (i.e., *lab3sol.py* file) and the output file (i.e., *lab3out.txt*) **through Brightspace.** The file *lab3sol.py* should contain the code for both PART 1 and PART 2.